

Is browsing the major factor of silver fir decline in the Vosges Mountains of France?

Patricia Heuze^{a,*}, Annik Schnitzler^a, François Klein^{b,1}

^a Université de Metz, Laboratoire Biodiversité et Fonctionnement des écosystèmes, Equipe de Phytoécologie,
Campus Bridoux, Avenue du Général Delestraint, 57070 Metz Cedex, France

^b Office National de la Chasse et de la Faune Sauvage, Centre National d'Etude et de Recherche Appliquée Sanglier-Cervidés,
1 place Exelmans, 55 000 Bar le Duc, France

Received 22 March 2005; received in revised form 2 June 2005; accepted 7 June 2005

Abstract

For many years, browsing wild ungulates have been considered the major cause for the decline of silver fir (*Abies alba*) in Europe. To better understand the dynamics of silver fir–beech forests hosting important deer populations, we studied forest renewal using a synchronistic approach, i.e. by comparing similar stands of different ages. Regeneration in a series of forests, where seedlings (<150 cm) were submitted to an increasing browsing pressure, was examined initially. We then analysed young stands (>2 m), which have grown in similar ecological conditions and also under increasing browsing. Published results on regeneration have identified three key factors as modulators of the effects of deer browsing on the composition of regeneration: understory abundance, stand composition and orientation to sun. This paper deals with the second part of the study. It examines the influence, over time, of the three environmental characteristics, identified at regeneration stage, by analysing the composition and the structure of young stands grown under comparable conditions. A total of 152 plots of 150 m² each were selected, spread over 30 km² of an ancient game reserve and core analysis was used to identify whether trees were under 150 cm when browsing became noticeable. Our results confirmed the critical role of stand composition, orientation to sun and understory—the latter being far predominant. Indeed, previous stand composition (i.e. seed trees at the time of regeneration) and orientation to sun did not explain current young stand composition alone, but when a south-exposed slope was previously occupied by mature spruce, then young spruce trees were frequent. Understory clearly played a positive role in the conservation of fir, despite increased browsing by deer. In fact, fir was abundant in the young stands when understory above seedlings was important at the time of regeneration, but when understory was poor, spruce became the dominant species and fir was seldom observed. Overall, silver fir was often the dominant species among young stands, despite a progressive increase in spruce numbers. The dominance of fir may be explained by the particular silvicultural history of the studied forests. However, in the future, the expansion of spruce, favoured by deer foraging on fir and by even-aged silviculture, may lead to localized failure of fir regeneration. To promote

* Corresponding author. Tel.: +33 3 87 37 84 24; fax: +33 3 87 37 84 25.

E-mail address: heuze@hotmail.com (P. Heuze).

¹ Tel.: +33 3 29 79 68 79.

silver fir–beech forest renewal, forestry practices should change towards a system more closely resembling natural forest dynamics.

© 2005 Elsevier B.V. All rights reserved.

Keywords: Silver fir–beech forests; Spruce; Silver fir decline; Deer browsing; Understory cover; Silviculture; Wildlife management

1. Introduction

Forests in the Vosges Mountains of France have undergone significant changes over the last 200 years. Firstly, silver fir–beech forests are the main formations found on the mountain level of the Vosges (Ellenberg, 1988; Ozenda, 1994), but due to forestry management practices, conifers now dominate the taxonomic composition. In fact, by the beginning of the 19th century, an even-aged silviculture had been established. Native deciduous trees were replaced by a few species of conifers, primarily spruce (*Picea abies*), which were favoured as fast growing and easy to establish and manage. Secondly, since the 18th century, wildlife management was involved in the extinction of large carnivores, such as bears (*Ursus arctos*), lynx (*Lynx lynx*) and wolves (*Canis lupus*). Following the French Revolution in 1789 and the subsequent lifting of the royal ban on hunting, herbivore populations were also significantly lowered by heavy culling. During the last decades, the reintroduction and protection of ungulate species, the creation of game reserves, additional feeding and restrictive hunting legislation have contributed to the rapid rise in wild large herbivore populations. These changes in forest and wildlife management significantly modified the interactions between plants and animals, especially between deer species *Cervus elaphus* and *Capreolus capreolus* and silver fir (*Abies alba*). Indeed, it would seem that increased browsing of silver fir by deer in winter is leading to the replacement of fir by spruce, which is not browsed heavily in the Vosges.

Large herbivores can actually have a significant influence on the composition of forest regeneration in favour of the less-browsed species (Gill, 1992a,b; Putman, 1994; Motta, 1996; Danell et al., 2003; Husheer et al., 2003; Rooney and Waller, 2003). However, in a managed system, silvicultural choices are also strongly involved in driving regeneration dynamics, sometimes in opposition to natural pro-

cesses. The natural silver fir–beech forest commonly regenerates through infrequent small gaps created by the fall or death of senescent trees (Korpel, 1982; Mazur, 1989; Boncina, 2000; Schnitzler et al., 2004). In such a regeneration pattern, only individuals from slow-growing, shade-tolerant species, e.g. fir, beech (*Fagus sylvatica*) or yew (*Taxus baccata*)—survive and grow (Runkle, 1981; Oldeman, 1990; Brzeziecki and Kienast, 1994). Regeneration is more rapid in large gaps that occur on steep slopes or in places exposed to wind. In these more sun-lit parts of the forest, other tree species of the silver fir–beech forest flourish, such as spruce, silver birch (*Betula pendula*), ash (*Fraxinus excelsior*), sycamore (*Acer pseudoplatanus*), rowan (*Sorbus aucuparia*), among other (Peters and Poulson, 1994; Diaci, 2002). The recurrent supply of light, following cutting and removal of understory by even-aged silviculture, accelerates establishment and growth of spruce (Aussenac and Ducrey, 1978; Grassi and Bagnaresi, 2001), and consequently, its expansion throughout the silver fir–beech forest. Forestry Department data indicate that spruce is increasingly common in adult stands of silver fir–beech forests of the Vosges, and unrelated to fir browsing as deer populations started to increase about 40 years ago, i.e. too recently to be involved.

To better understand the dynamics of the silver fir–beech forest subjected to the double influence of large herbivores and foresters, we studied forest renewal by using a synchronistic approach, i.e. by comparing stands of different ages, which have grown in similar conditions. In the first part of the study, Heuzé et al. (2005) have highlighted, among several environmental characteristics, the role of three parameters in modulating the effect of browsing on seedling composition. Firstly, on slopes with southern exposures, many fir seedlings did not reach heights above 30 cm and spruce developed; this was related to a high browsing intensity on fir because deer are attracted by sunny places in winter. Secondly, spruce expansion was facilitated, at the detriment of fir, by its presence

in nearby mature stands and this was not related to a particularly high level of browsing on fir. Lastly, the presence of foliage from the understory above seedlings (corresponding to the percentage of the plot surface covered by foliage from either small trees, shrubs, seedlings more than 1.5 m tall, or low branches of higher trees) was the only factor found to benefit fir by limiting spruce development and reducing browsing intensity.

Stand composition and understory abundance are linked to silvicultural choices. Appropriate management of vegetation may thus limit browsing effects and spruce expansion, but it relies on understanding the role, over time, of the manipulated parameters. Therefore, this study was carried out in forests similar to those studied by Heuzé et al. (2005) to analyse the composition and the structure of young stands, which have grown over 40 years of increased browsing pressure, under various conditions of understory, stand composition and orientation to sun.

2. Study area

The Vosges Mountains are located in northeastern France (47°40'N to 48°50'N, 6°40'E to 7°30'E), reaching an elevation of 1424 m at their highest point. The research was carried out in the southern part of the Vosges, described by Cachan (1974) and Oberti (1990). The principal soil type is acid brown on Hercynian granites. The climate is oceanic. Mean annual precipitation ranges from 1000 to 2000 mm. Average annual temperatures vary from 4 °C at 1200 m to 8 °C at 400 m, with average variations of 17 °C between summer and winter. Snow is frequent on the summits during the period from November to April, and often remains at the highest elevations until May. The crest is covered by subalpine heath–grassland, below which there are beech forests. Silver fir–beech forests are the main formation at mountain level (500–950 m), but spruce has expanded into these forests owing to silviculture practices and it is now dominant, either pure or mixed with fir, over broad areas (Husson, 1988). The ground flora, typical of acid soils at these altitudes, has abundant *Vaccinium myrtillus* L., *Deschampsia flexuosa* L., *Luzula luzuloides* Lam., *Luzula sylvatica* and *Festuca altissima* All.

The study site was selected in the natural range of the silver fir–beech forest, within a 30 km² area. It includes the forest of Haute Meurthe and parts of the neighbouring forests of Ban sur Meurthe and Gérardmer, where the red deer population started to increase in the early 1960s following the creation of the 'Belbriette' game reserve. Data from the Forestry Department indicate that in 1960 mature stands of these forests were composed of 70–80% fir, <10% beech and 15–20% spruce. Nowadays, mature spruces are twice as numerous as in 1960 and represent 30–50% of the mature trees. In about 1970, foresters observed a lack of fir seedlings resulting from browsing, correlated with an increase in naturally regenerating spruce. In 1978, severe damages to trees (browsing, fraying and bark stripping) were reported, and the reserve was closed. Culling of animals was begun, but with small quotas, so that the population of red deer continued to increase (Heim, 1984). The density of red deer inside the reserve was estimated, by night counting carried out by both hunters and foresters, at 4.5 animals per 100 ha in the early 1980s, 13 in 1995 and 7 in 2000, following strong hunting pressure by the end of the 1990s. Forest and game managers agree that the carrying capacity for red deer in these forests is two or three red deer per 100 ha. Roe deer were quite abundant before the reserve was established, but their population decreased as the red deer population rose. The estimated five roe deer per 100 ha in the early 1980s dropped to 1.7–1.9 between 1995 and 2000. Nevertheless, the simultaneous presence of roe and red deer for decades prior to the present-day abundance in the latter species precludes a clear differentiation in the role of each species.

3. Methods

3.1. Sampling and data collection

A selection of areas was first made based on Forestry Department data. Areas were chosen that regenerated (by seed tree or shelterwood cuttings) between 1960 and 1980 and where mature stands, at the time of regeneration, were composed mainly of fir and/or spruce. In each area, 10 m × 15 m plots were sampled on transects running along the hillside every

50 m in elevation, with a 200 m distance between plots. Cores analysis was carried out in order to ascertain whether trees were less than 150 cm in 1970, i.e. accessible to red deer when selective browsing of fir became conspicuous. Cores were sampled on the 10 firs and spruces having the greatest number of whorl above 150 cm, which were assumed to be the oldest trees in the plot. Cores were collected at 150 cm height, so that those showing less than 30 rings revealed that the tree was less than 150 cm in 1970. When all the cores displayed more than 30 rings, denoting that all the trees were higher than 150 cm in 1970, additional cores were collected on 10 other individuals and the plot was studied only if these additional trees all showed less than 30 rings. Trees higher than 150 cm in 1970 were considered as part of the understory covering the seedlings in 1970.

A total of 152 plots were studied in summer 2001, spread over the 30 km² of the ancient game reserve. On average, plots contained 41 trees (S.D. 17); trees measured from 2 to 10 m high and diameter at breast height ranged from a few to 20 cm. Nevertheless, owing to the even-aged structure of the stands, heights and diameters were quite homogenous inside a plot.

The species of each tree of the plots was noted (fir, spruce, beech, other) as well as its social status. Indeed, when fir seedlings have grown protected from browsing by the crown of higher trees, it should be abundant in the dominated layer of the young stand. Three levels of status were defined: (i) dominant: the crown of the tree extends above the canopy layer, (ii) co-dominant: the crown forms the main canopy layer, (iii) dominated: the crown is entirely below the canopy or reaches only to the lower part.

The number of stumps was recorded, accounting for past silvicultural operations, such as thinning or removal of understory. Lastly, orientation to sun was determined using a compass.

3.2. Statistical analysis

Data treatment was performed using the software Statistica 6.0. Correspondence analysis and hierarchical classification were used to describe the young stands. These techniques aimed at defining types of stands by grouping plots, which exhibited the same distribution of trees from the three species, into three status classes (Legendre and Legendre, 1998).

To organise the dataset around independent axes, correspondence analysis was performed on a table composed of 12 columns (3 status $\times$ 4 species) and 152 lines. To identify types of stands, the coordinates of plots on the first axis, explaining up to 80% of total variation, were then analysed in the hierarchical classification, using Euclidian distance measurements and the Ward method of aggregation. Chi-square analysis was then used to test the distribution homogeneity of stand types for the various modalities of environmental descriptors. For quantitative data, i.e. the percentage of trees taller than 150 cm in 1970, a discriminant analysis using the classification and regression trees method (Breiman et al., 1984) was performed to define the threshold values, where observed distribution varied significantly from theoretical distribution. When data did not show a Gaussian distribution, non-parametric statistics were used: Mann–Whitney *U*-test for comparisons of numbers and Spearman's *R* of for correlations.

4. Results

An overview of the distribution of the plots over the environmental conditions is given Table 1.

4.1. Influence of past silvicultural operations on the accuracy of present-day data

Because of forestry works, such as thinning or removal of understory, our data may not fit the situation 30 years ago and results may be distorted. Firstly, the proportion of trees covering seedlings in 1970 was estimated from the observation of trees present today, but some may have been cut in the intervening time and stumps should also be accounted for past understory. The number of stumps was actually negatively correlated with the percentage of trees taller than 150 cm in 1970 (Spearman's $R = -0.25$, $t = -3.09$, $P < 0.01$), but the relationship was no longer significant when analysing only the plots containing less than 16 stumps, which represented 87% of the sample (Spearman's $R = -0.15$, $t = -1.64$, $P > 0.1$). The assessment of the cover in 1970 through core analysis was, therefore, little influenced by foresters' interventions. Secondly, apprehensive of seeing spruce replacing silver fir,

Table 1
Distribution of each young stand in the modalities of environmental conditions

Young stands	Orientation		Seed trees composition			Understory coverage (%)		
	South <i>N</i> (%)	n + w + e <i>N</i> (%)	Fir <i>N</i> (%)	Spruce <i>N</i> (%)	Fir–spruce <i>N</i> (%)	[0–4] <i>N</i> (%)	[4–16] <i>N</i> (%)	[16–28] <i>N</i> (%)
FIR	38 (35)	70 (65)	62 (58)	22 (20)	24 (22)	27 (25)	61 (56)	20 (19)**
SPRUCE	12 (50)*	12 (50)	13 (54)	5 (21)	6 (25)	14 (58)**	9 (38)	1 (4)
FIR&SPRUCE	7 (58)*	5 (42)	9 (75)	0	3 (25)	5 (42)	7 (58)	0
FIR&BEECH	2 (25)	6 (75)	6 (75)	0	2 (25)	4 (50)	4 (50)	0
Total sample	59 (39)	93 (61)	90 (59)	27 (18)	35 (23)	50 (33)	81 (57)	21 (10)

N (%) corresponds to the number of plot in each modality and to the distribution (%) of the plots per young stand type in each environmental condition. A significant difference indicates that the type of young stand was more often observed in the modality that is mentioned. See Section 4 for detailed explanations.

* $P = 0.06$ (χ^2 -test).

** $P < 0.02$ (χ^2 -test).

foresters attempted to favour fir by spruce removal (personal communication from foresters), so that the relative abundance of fir should increase with the number of stumps. The correlation was not significant (Spearman's $R = 0.16$, $t = 1.93$, $P > 0.05$), so we deemed that forestry intervention poorly influenced the composition of present young stands.

4.2. Young stand composition and structure

Four types of young stands were defined based on tree composition (Fig. 1). Most of the plots were covered with pure stands dominated by fir (FIR type: 71% of the sample) or spruce (SPRUCE: 16% of the sample), mixed stands being poorly represented (8% of the sample for FIR&SPRUCE and 5% for FIR&BEECH). It was not possible to distinguish stands according to their structure, as the dominance or the mixture of species was constant among social statuses. There was, therefore, no evidence for the existence of a situation where dominated firs have grown under the canopy of spruce, as expected. On the other hand, the dominated layer of the FIR&BEECH type was composed of 21.4% of the total number of fir and only 11.2% of beech ($U = 9.5$, $P < 0.02$).

4.3. Influence of seed tree composition and orientation to sun on young stand composition

The composition of young stands was not significantly related either to the dominant species among seed trees at the time of regeneration nor to orientation to sun, but stands containing spruce

(SPRUCE and FIR&SPRUCE) tended to be more frequent on south-exposed slopes (Table 1; $\chi^2 = 3.49$, d.f. = 1, $P = 0.06$). Nevertheless, when these environmental characteristics were studied together, they did explain the composition of young stands. Indeed, among the young stands located on areas which previously contained spruce, SPRUCE and FIR&SPRUCE stands were more frequent on south-exposed slopes compared with FIR type (Fig. 2; $\chi^2 = 8.32$, d.f. = 1, $P < 0.001$).

4.4. Influence of past understory on young stand composition

Thanks to core analysis, the extent of understory covering seedlings in 1970 was assessed using the percentage of trees identified as taller than 150 cm in 1970 (number of trees > 150 cm in 1970/number of total trees on the plot). Discriminant analysis showed that when the estimated past understory was composed of less than 4% of the trees, the SPRUCE type was frequent and FIR type was seldom observed. Contrarily, FIR type was abundant when the understory constituted more than 16% of the trees; all other types being absent or rare (Table 1; $\chi^2 = 15.27$, d.f. = 6, $P < 0.02$).

4.5. Evolution of the abundance of spruce among young stands

Core analysis elucidated the year when trees in a plot started to grow taller than 150 cm, allowing the classification of plots along a temporal axis. For this

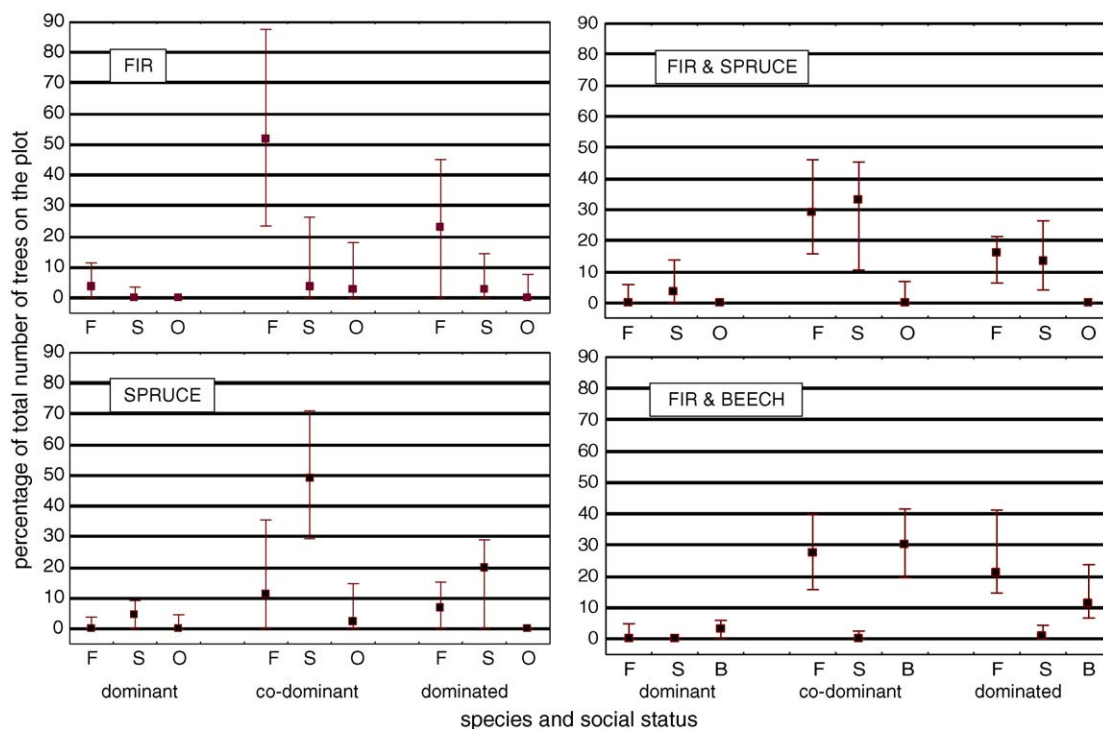


Fig. 1. Young stands composition and structure. Squares represent the median value and bars indicate the minimum and maximum values. F, fir; S, spruce; O, other (mainly beech); B, beech. Number of FIR stands = 108, number of SPRUCE stands = 24, number of FIR&SPRUCE stands = 12, number of FIR&BEECH stands = 8.

reason, the median number of rings was calculated for each plot and subtracted from 2001—the year of data collection. Spearman's R revealed an increase with time of the percentage of spruce in the young stands. The correlation was not striking, but highly significant (Fig. 3; $R = 0.37$, $t = 4.88$, $p < 0.001$).

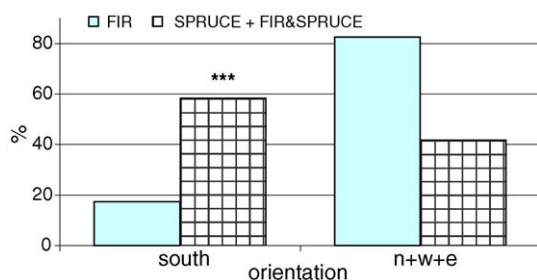


Fig. 2. Composition of the young stands replacing spruce containing stands related to orientation to sun. n + w + e: north + west + east. Number of south-exposed plots = 15, number of (n + w + e)-exposed plots = 43, number of FIR stands = 46, number of SPRUCE + FIR&SPRUCE stands = 12. χ^2 -test: *** $P < 0.001$.

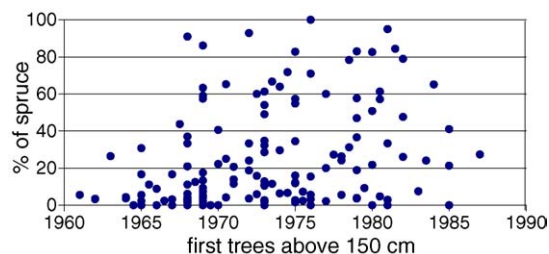


Fig. 3. Evolution of the abundance of spruce among young stands. The plots on the left side of figure were covered by understory in 1970.

5. Discussion

5.1. Lessons from the past

As shown by the importance of FIR stands in our sample (71%), increased heavy winter browsing of fir over the last decades has not lead to a significant replacement of fir by spruce among the young trees of

the silver beech forest, even if spruce is actually becoming more numerous. The current feature of young stands results from the biological and ecological attributes of the studied species, which have grown in a particular silvicultural context. Indeed, forestry activities in the Vosges Mountains decreased from the end of the 19th to the middle of the 20th century, so that mature stands closed-up while growing old. Owing to its low light requirements, silver fir could grow in these shaded stands, but spruce was excluded and developed only in more sun-lit places with poor understory. This was supported by several fir cores, which showed a great number of small rings, hinting that tree growth was suppressed for many years. The role of understory on limiting spruce development was clearly confirmed. Indeed, the plots which were the more covered by foliage from trees higher than 150 cm at the time of regeneration were now mostly occupied by young silver firs. Fir was, therefore, the major regenerating species in coniferous stands around 1960. The amount of seeds, provided by mature firs present at that time, allowed massive establishment of seedlings, even under spruce canopies. Fir development was also facilitated by the absence of strong competitors, as beech was invariably cut by foresters and spruce was limited by the rarity of seed producers, in addition to unfavourable light conditions. Fir was, therefore, already established in places when the Fond Forestier National and the Office National des Forêts, commissioned to rejuvenate forested areas, were created in 1946 and 1966, respectively. The wide-scale opening of forests, which followed favoured natural regeneration of spruce. It also contributed to the rise of ungulate populations, initiated by wildlife management measures, such as game reserves, by increasing food resources through the creation of numerous open areas (regenerating stands, clear cuts, plantations, etc.), roads and edges. Increased foraging by deer, in addition to competition with spruce, constituted an important threat that fir had, and still has, to overcome to maintain itself in the silver fir–beech forest.

Because fir scarcely endures herbivory (Louda et al., 1990; McLaren, 1996; Ayres et al., 2004), its ability to establish and survive in shaded conditions is a real advantage in the presence of deer. Areas covered by understory are shelters of sort, where seedlings can escape heavy browsing and constitute what Milchunas

and Noy-Meir (2002) called *biotic refuges* (i.e. other plants that protect a target plant). Similar processes have been reported by several authors (Welch et al., 1991; Proudhon et al., 1995; Reimoser and Gossow, 1996; Schreiner et al., 1996; Pietrzykowski et al., 2003). Other than protection from browsing, understory also improves ecological conditions for fir seedlings. It prevents over-exposed to sun and uprooting when soil is washed away during strong rains, two important causes of fir mortality (Tan, 1987), and it reduces the risk of spring frosts, which affect fir more than spruce (Aussenac, 2000; Schweingruber and Müller, 1992). On the other hand, understory trees, mainly beech and spruce, are also potential competitors with silver fir. It seems that the protective advantage of spruce at the regeneration stage does not outweigh the cost of competition over time. Indeed, at the early stage of regeneration, spruce may overtop the browsed fir seedlings (Ammer, 1996; Motta, 1999; Heuzé et al., 2005) and we expected firs would then grow under the crown of spruce. The absence of fir-dominated young stands suggests that fir can barely survive under spruce, probably because too little light passes through the dense crown of spruce throughout the year (Robakowski et al., 2004) as well as too little water. Nevertheless, the two species are able to grow jointly, as shown by the mixture of young spruce and fir. When silver fir is associated with beech, the dominated firs represent up to 42% of trees (versus a maximum of 22% in the fir and spruce stands). The deep roots of young beech (Riedacker, 1981) may permit other plants to compete for soil nutrients and establish new individuals easier than the shallow-rooting system of spruce (Duchaufour, 1953). Nevertheless, to ascertain the role of beech on the survival and growth of fir in a browsing context, a more detailed study of understory characteristics is necessary, encompassing all broadleaved species of the silver fir–beech forest. Moreover, if tall vegetation is useful for fir preservation, this needs to be weighed against the potential damaging competitive effects of such vegetation.

5.2. Outlining the future

Our results argue that the percentage increase of spruce among young trees is a consequence of the combined action of unsuitable silviculture practices

and fir browsing by deer. Therefore, a reduction in the deer population is not the solution per se (Güthorl, 1994; Putman et al., 1996; Coomes et al., 2003). Furthermore, analysis based on isolated components are ecologically deficient and an overall ecosystem analysis is necessary to provide a rigorous ecological basis for managing silver fir–beech forest and thwarting spruce expansion.

Owing to our current state of knowledge and the unpredictability of global change (climatic variability, heavy storms, return of large predators, epidemic disease of deer or trees, etc.), it is difficult to predict the long-term influence of browsing ungulates on the ecosystems. Nevertheless, it is possible, through appropriate measures, to reduce tree susceptibility to deer foraging. Our results have shown that on south-exposed slopes with mature spruce and poor understory, renewal of fir stands cannot be guaranteed. It is obvious, then, that silviculture should try to maintain fir in the canopy at a maximum, favour understory development and leave at least part of the existing cover intact during the regeneration phase. Particular attention should be given to the south-exposed areas, which are attractive for herbivores in winter (Lowe, 1966; Staines, 1977; Coughenour, 1991). Thus, natural regeneration of spruce can be limited, to the benefit of fir, by regenerating stands in small gaps (Ellenberg, 1988; Grassi and Bagnaresi, 2001; Paluch, 2004). This regime, when repeated frequently, supports understory and ground vegetation (as an alternative food source) (Lakière et al., 2001). Autochthonous broadleaves, such as ash (*F. excelsior*), sycamore (*A. pseudoplatanus*), rowan (*S. aucuparia*), elm (*Ulmus glabra*), elder (*Sambucus nigra*), alpine elder (*Sambucus racemosa*), holly (*Ilex aquifolium*) should also be encouraged to grow, as they may constitute an efficient understory. They are less competitive than spruce and improve the availability of alternative forage (on their own and by allowing the development of ground vegetation), thus limiting the risk of browsing on fir (Eiberle and Bucher, 1989; Guibert et al., 1992).

These measures fit the pattern of uneven-aged management, which is closer to the natural functioning of silver fir–beech forests, and could help increase quality and quantity of food available for herbivores (Jedrzejewska et al., 1994; Klein et al., 1999; Lakière et al., 2001). Therefore, silver fir–beech forests of the

Vosges Mountains would be more resistant to deer-foraging, but also to many biotic and abiotic threats, such as fungi or insect attack, wind/soil erosion or snow damage (Korpel, 1982; Brang, 2001; Spiecker, 2003). Nevertheless, the development of understory and broadleaves, as well as the spreading of fir seedlings in small clearings, may be impeded by local high densities of large herbivores. A temporary reduction in the deer population may be necessary to initiate forest transformation, but in the long term, numerical and spatial fluctuation in animal populations need to be considered, as they are part of the natural dynamics of forest ecosystems. As underlined by Senn and Suter (2003), wildlife management aims at stable ungulate populations without significant fluctuations, and recolonising large predators will not be permitted. Nonetheless, for plants, a period of low ungulate numbers would contribute to temporary relief from heavy browsing. In addition, predation by large wild carnivores is an important element in determining spatial distribution of ungulates and their impact on regeneration. For example, ungulates avoid areas where wolf packs hunt, and the negative impact on heavily browsed species is lowered (Hoskinson and Mech, 1976; Ripple and Beschta, 2004).

6. Conclusion

The western limit of spruce has been moved considerable to the west by intensive forest management, which has, coupled with the rise in ungulate numbers, significantly modified silver fir–beech forest dynamics in the Vosges. Spruce could be controlled, to the benefit of fir, by more natural silviculture practices. Nevertheless, spruce is now well established and will almost certainly retain its pre-eminent status. Therefore, further study is needed to better understand the interactions between the main tree species (fir, beech and spruce) and the involvement of browsing by large herbivores.

Acknowledgements

This study was undertaken thanks to financial and technical support received from the French Office National de la Chasse et de la Faune Sauvage, Parc

Naturel Régional des Ballons des Vosges, Direction Régionale de l'Environnement and Office National des Forêts.

References

- Ammer, C., 1996. Impact of ungulates on structure and dynamics of natural regeneration of mixed mountain forests in the Bavarian Alps. *For. Ecol. Manage.* 88, 43–53.
- Aussenac, G., Ducrey, M., 1978. Etude de la croissance de quelques espèces forestières cultivées à différents niveaux d'éclaircissement et d'alimentation hydrique. In: 106^{ème} Congrès des Sociétés Savantes, Nancy 1, 105–117.
- Aussenac, G., 2000. Interactions between forest stands and microclimate: ecophysiological aspects and consequences for silviculture. *Ann. For. Sci.* 57, 287–301.
- Ayres, E., Heath, J., Possell, M., Black, H.I.J., Kerstiens, G., Bardgett, R.D., 2004. Tree physiological responses to above-ground herbivory directly modify below-ground processes of soil carbon and nitrogen cycling. *Ecol. Lett.* 7, 469–479.
- Brang, P., 2001. Resistance and elasticity: promising concepts for the management of protection forests in the European Alps. *For. Ecol. Manage.* 145, 107–119.
- Boncina, A., 2000. Comparison of structure and biodiversity in the Rajhenav virgin forest remnant and managed forest in the Dinaric region of Slovenia. *Global Ecol. Biogeogr.* 9, 201–211.
- Breiman, L., Friedman, J.H., Olshen, R.A., Stone, C.J., 1984. *Classification and Regression Trees*. Wadsworth & Brooks, Monterey.
- Brzeziecki, B., Kienast, F., 1994. Classifying the life-history strategies of trees on the basis of the Grimian model. *For. Ecol. Manage.* 69, 167–187.
- Cachan, P., 1974. Etude bioclimatique du massif vosgien. *Bull. E.N.S.A.I.A.* 16, 3–33.
- Coomes, D.A., Allen, R.B., Forsyth, D.M., Lee, W.G., 2003. Factors preventing the recovery of New Zealand forests following control of invasive deer. *Conserv. Biol.* 17, 450–459.
- Coughenour, M.B., 1991. Spatial components of plant-herbivore interactions in pastoral, ranching, and native ungulate ecosystems. *J. Range Manage.* 44, 530–542.
- Danell, K., Bergström, R., Edenius, L., Ericsson, G., 2003. Ungulates as drivers of tree population dynamics at module and genet levels. *For. Ecol. Manage.* 181, 67–76.
- Diaci, J., 2002. Regeneration dynamics in a Norway spruce plantation on a silver fir-beech forest site in the Slovenian Alps. *For. Ecol. Manage.* 161, 27–38.
- Duchaufour, P., 1953. Régénération de l'épicéa et pédologie. *Revue forestière française* 4, 257–268.
- Eiberle, K., Bucher, H., 1989. Interdependance between browsing of different tree species in a selection forest region. *Z. Jagdwiss.* 35, 235–244.
- Ellenberg, H., 1988. *Vegetation Ecology of Central Europe*. Cambridge University Press.
- Gill, R.M.A., 1992a. A review of damage by mammals in north temperate forests: 1. Deer. *Forestry* 65, 145–169.
- Gill, R.M.A., 1992b. A review of damage by mammals in north temperate forests: 3. Impact on trees and forests. *Forestry* 65, 363–388.
- Grassi, G., Bagnaresi, U., 2001. Foliar morphological and physiological plasticity in *Picea abies* and *Abies alba* saplings along a natural light gradient. *Tree Physiol.* 21, 959–967.
- Guibert, B., Maizeret, C., Ballon, P., Montes, E., 1992. Influences of forest management on Roe deer populations in the Landes of Gascony. In: *Proceedings of the international symposium "Ongulés/Ungulates"*, Toulouse, France, pp. 617–620.
- Güthorl, V., 1994. Roe deer *Capreolus capreolus* population density and browsing pressure in a mixed-forest ecosystem. *Forest and wildlife... towards the 21st century*, 21st congress of the International Union of Game Biologists, Halifax, Canada, 1993, pp. 222–229.
- Heim, D., 1984. Gibier et sylviculture dans les forêts d'altitude de la subdivision de St. Dié-Fraize, Vosges. *Rapport ONF, St. Dié des Vosges*.
- Heuzé, P., Schnitzler, A., Klein, F., 2005. Consequences of increased deer browsing winter on silver fir and spruce regeneration in the Southern Vosges Mountains: implications for forest management. *Ann. For. Sci.* 62, 175–181.
- Hoskinson, R.L., Mech, D.L., 1976. White tailed deer migration and its role in wolf predation. *J. Wildlife Manage.* 43, 429–441.
- (Okarma, H., 1998. *Le loup en Europe*. Grands Espaces, France).
- Husheer, S.W., Coomes, D.A., Robertson, A.W., 2003. Long-term influence of introduced deer on the composition and structure of New Zealand *Nothofagus* forests. *For. Ecol. Manage.* 181, 99–117.
- Husson, J.P., 1988. Les forêts du département des Vosges, 1820–1945, le temps des transformations décisives. *Annales de la Société d'Emulation des Vosges* 85–94.
- Jedrzejewska, B., Okarma, H., Jedrzejewski, W., Milkowski, L., 1994. Effects of exploitation and protection on forest structure, ungulate density and wolf predation in Białowieża Primaverla Forest Poland. *J. Appl. Ecol.* 31, 664–676.
- Klein, F., Heuzé, P., Brosier, P., 1999. Biodiversité et gestion forestière, l'exemple des Vosges du Nord: étude des forêts en tant qu'habitat des cervidés. *Annales Scientifiques de la Réserve de la Biosphère des Vosges du nord* 7, 103–117.
- Korpel, S., 1982. Degree of equilibrium and dynamical changes of the forest on example of natural forests of Slovakia. *Acta Facultatis Forestalis Zvolen XXIV*, 9–31.
- Lakière, V., Normant, P., Ballon, P., 2001. Disponibilités alimentaires pour le chevreuil (*Capreolus capreolus*) dans deux sapinières régulière et jardinée du second plateau du Jura. *Revue Forestière Française* 5, 511–526.
- Legendre, P., Legendre, L., 1998. *Numerical Ecology*. Elsevier Science, Amsterdam.
- Louda, S.M., Keeler, K.H., Holt, R.D., 1990. Herbivore influences on plant performance and competitive interactions. In: Grace, J.B., Tilman, D. (Eds.), *Perspectives on Plant Competition*. Academic press, pp. 413–444.
- Lowe, V.P.W., 1966. Observations on the dispersal of red deer on Rhum. *Symp. Zool. Soc. Lond.* 18, 211–228.

- Mazur, M., 1989. Structure and dynamics of silver fir (*Abies alba* Mill.) in forest communities of the Swietokrzyski National Park. II. Population dynamics. *Acta Soc. Bot. Pol.* 58, 409–422.
- McLaren, B.E., 1996. Plant specific response to herbivory: simulated browsing of suppressed balsam fir on Isle Royale. *Ecology* 77, 228–235.
- Milchunas, D.G., Noy-Meir, I., 2002. Grazing refuges, external avoidance of herbivory and plant diversity. *Oikos* 99, 113–130.
- Motta, R., 1996. Impact of wild ungulates on forest regeneration and tree composition of mountain forests in the Western Italian Alps. *For. Ecol. Manage.* 88, 93–98.
- Motta, R., 1999. Wild ungulate browsing, natural regeneration and silviculture in the Italian Alps. *J. Sustain. For.* 8, 35–53.
- Oberti, D., 1990. Catalogue des stations forestières des Vosges alsaciennes. Fascicule1: les Vosges cristallines. Université de Bourgogne, E.N.G.R.E.F., Nancy.
- Oldeman, R.A.A., 1990. *Forests: Elements of Silvology*. Springer-Verlag, Berlin.
- Ozenda, P., 1994. *Végétation du Continent Européen*. Delachaux et Niestlé, Lausanne, Paris.
- Paluch, J.G., 2004. The influence of the spatial pattern of trees on forest floor vegetation and silver fir next term (*Abies alba* Mill.) regeneration in uneven-aged forests. *For. Ecol. Manage.* 205, 283–298.
- Peters, R., Poulson, T.L., 1994. Stem growth and canopy dynamics in a world-wide range of *Fagus sylvatica* forests. *J. Veg. Sci.* 5, 421–432.
- Pietrzykowski, E., McArthur, C., Fitzgerald, H., Goodwin, A.N., 2003. Influence of patch characteristics on browsing of tree seedlings by mammalian herbivores. *Ecology* 40, 458–469.
- Proudhon, P., Guibert, G., Klein, F., Demolis, C., 1995. Rôles et fonctions des différents types de peuplements forestiers en période hivernale vis-à-vis des grands cervidés. Cas du massif forestier d'Is-sur-Tille. *Revue Forestière Française XLVII*, 137–150.
- Putman, R.J., 1994. Effects of grazing and browsing by mammals on woodlands. *Brit. Wildlife* 5, 205–213.
- Putman, R.J., Langbein, J., Hewison, A.J., Sharma, S.K., 1996. Relative roles of density-dependent and density-independent factors in population dynamics of British deer. *Mammal Rev.* 26, 81–101.
- Reimoser, F., Gossow, H., 1996. Impact of ungulates on forest vegetation and its dependence on the silvicultural system. *For. Ecol. Manage.* 88, 107–119.
- Riedacker, A., 1981. In: Le Tacon, F., Nepveu, G., Pardé, J., Perrin, R., Timbal, J., Le Hêtre, (Eds.), *Physiologie de la Croissance, Croissance Aérienne et Souterraine*. Lavoisier, Paris.
- Ripple, W.J., Beschta, R.L., 2004. Wolves, elk, willows, and trophic cascades in the upper Gallatin Range of Southwestern Montana USA. *For. Ecol. Manage.* 200, 161–181.
- Robakowski, P., Wykab, T., Samardakiewicz, S., Kierzkowskia, D., 2004. Growth, photosynthesis, and needle structure of silver fir (*Abies alba* Mill.) seedlings under different canopies. *For. Ecol. Manage.* 201, 211–227.
- Rooney, T.P., Waller, D.M., 2003. Direct and indirect effects of white-tailed deer in forest ecosystem. *For. Ecol. Manage.* 181, 165–176.
- Runkle, J.R., 1981. Gap regeneration in some old-growth forests of the eastern United States. *Ecology* 62, 1041–1051.
- Schnitzler, A., Closset, D., Gafta, D., Cristea, V., 2004. Dynamique des populations et mosaïque forestière en hêtraie-sapinière naturelle préservée. Une comparaison entre Vosges et Carpates. *Rev. Ecol. Terre Vie.* 59, 213–228.
- Schreiner, E.G., Krueger, K.A., Happe, P.J., Houston, D.B., 1996. Understory patch dynamics and ungulate herbivory in old-growth forests of Olympic National Park, Washington, DC. *Can. J. For. Res.* 26, 255–265.
- Senn, J., Suter, W., 2003. Ungulate browsing fir (*Abies alba*) in the Swiss Alps: beliefs in search of supporting data. *For. Ecol. Manage.* 181, 151–167.
- Schweingruber, F.H., Müller, M., 1992. Dendrochronological evidence in fir and spruce of alternating frosts in Switzerland. *Dendrochronologia* 10, 23–28.
- Spiecker, H., 2003. Silvicultural management in maintaining biodiversity and resistance of forests in Europe-temperate zone. *J. Environ. Manage.* 67, 55–65.
- Staines, B.W., 1977. Factors affecting the seasonal distribution of red deer (*Cervus elaphus*) at Glen Dye, north-east Scotland. *Ann. Appl. Biol.* 87, 495–512.
- Tan, B.S., 1987. Etude de l'influence de quelques facteurs écologiques sur les premiers stades de la régénération du sapin (*Abies alba*) à l'étage montagnard du Jura. *Revue d'Ecologie et de Biologie des Sols* 24, 623–635.
- Welch, D., Staines, B.W., Scott, D., French, D.D., Catt, D.C., 1991. Leader browsing by red and roe deer on young sitka spruce trees in western Scotland. I. Damage rates and the influence of habitat factors. *Forestry* 64, 61–82.